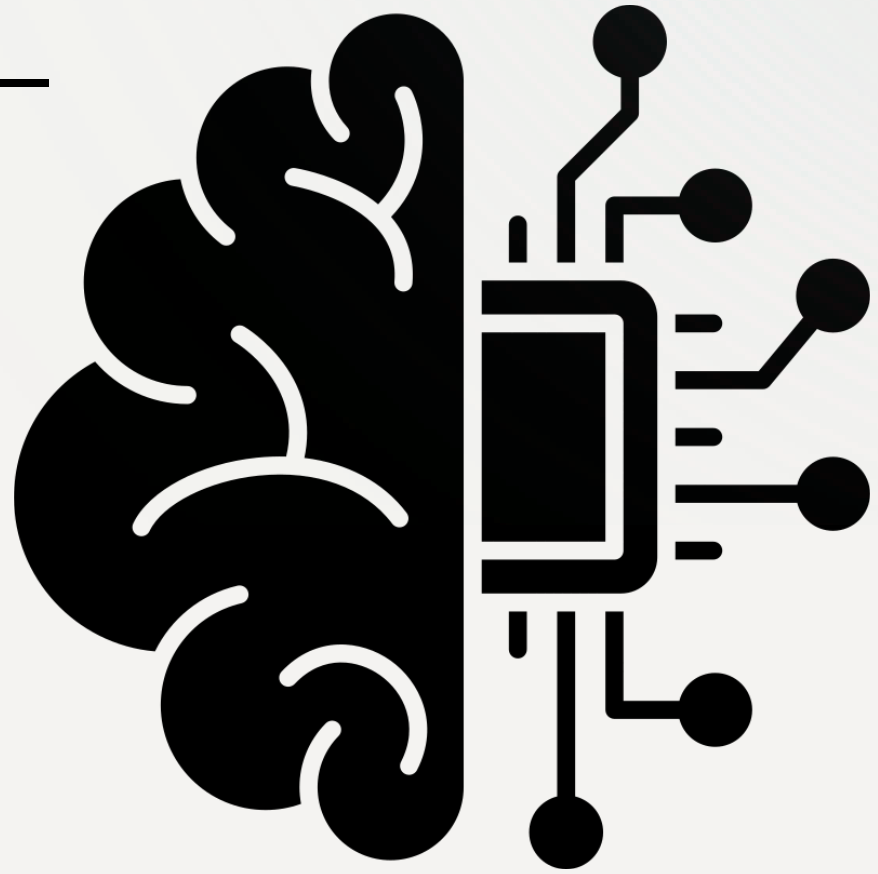
POLITECNICO MILANO 1863

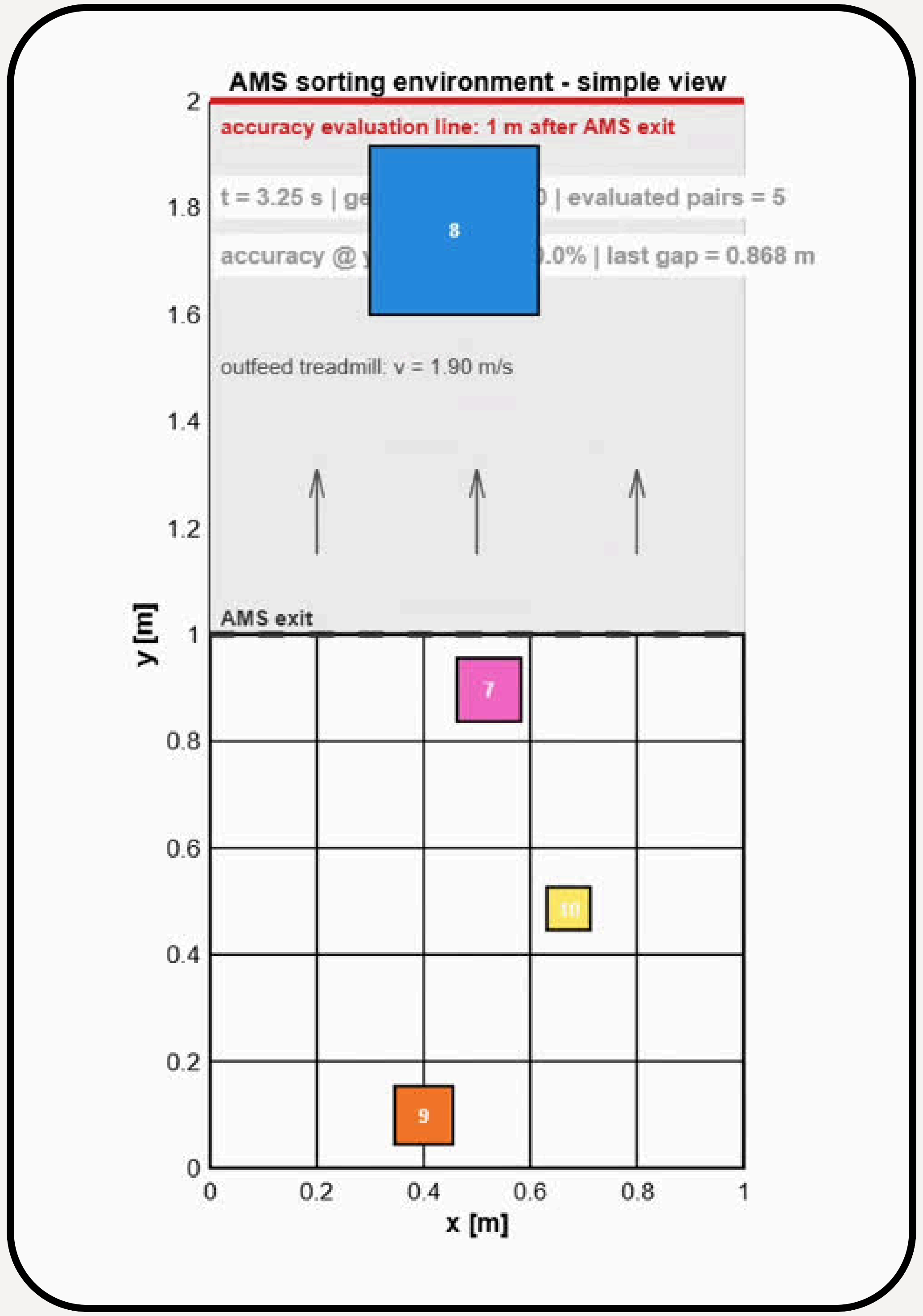
MACHINE LEARNING FOR MECHANICAL SYSTEMS
A.Y. 2025-2026

PROJECT PRESENTATION



Presented by: Yigiter Tuncer

RL MATLAB: PARCEL SINGULATION



CORE ENGINEERING FIXES AND DATA INFRASTRUCTURE

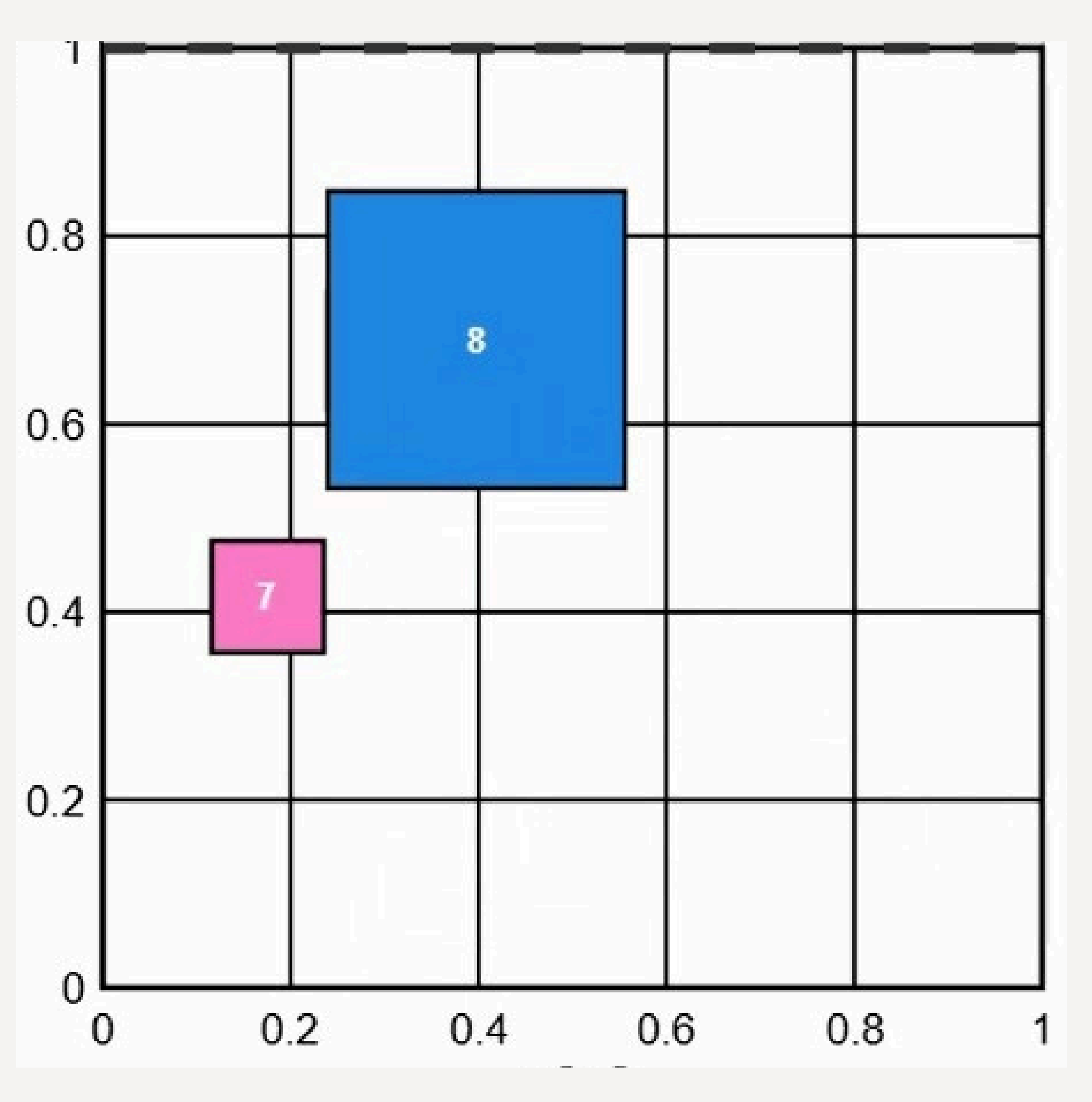
1) Dimensionality Bug Fix: Correction of the issue where upward velocity (vy) was displayed as m/step by dividing it by Δt to synchronize it with the actual m/s.

2) Spawn Logic Shift: Decision to use both time-based (continuous) generation and wave-based generation for different scenarios

3) Collision Tolerance: Matching the physics engine's 0.005 m thrust tolerance with the RL agent's 0.006 m penalty tolerance.

4) Decoupled Video Logging: Replaced heavy real-time and simple AMSvisualizer with a matrix logger (simulation_log.mat) and a standalone video_saver.m for smooth 30 FPS MP4 rendering.

5) Post-Training Plot Extraction: Developed RewardPlot.m to reconstruct and customize high-quality convergence plots directly from raw trainingStats.mat arrays, preventing 14-hour data loss.



REWARD DESIGN

- 10 different reward versions were tested across 15 training runs, iteratively solving problems observed in each run's behavior.

The "Bang-Bang" Jittering

LINEAR GRADIENT

Problem: Aggressive quadratic penalties (error^2*10) caused severe lateral vibrations.
Solution: Smoothed into linear gradients for stable deceleration.

The "Catch-22" Paradox

IS_BLOCKED FLAG

Problem: Speed limit penalty vs. Collision penalty created a deadlock.
Solution: Adaptive Speed Control (is_blocked flag) exempts blocked agents from speed limits.

Centering vs. Overtaking

PARABOLIC WEIGHT (0.3 + 1.7 proximity^2)

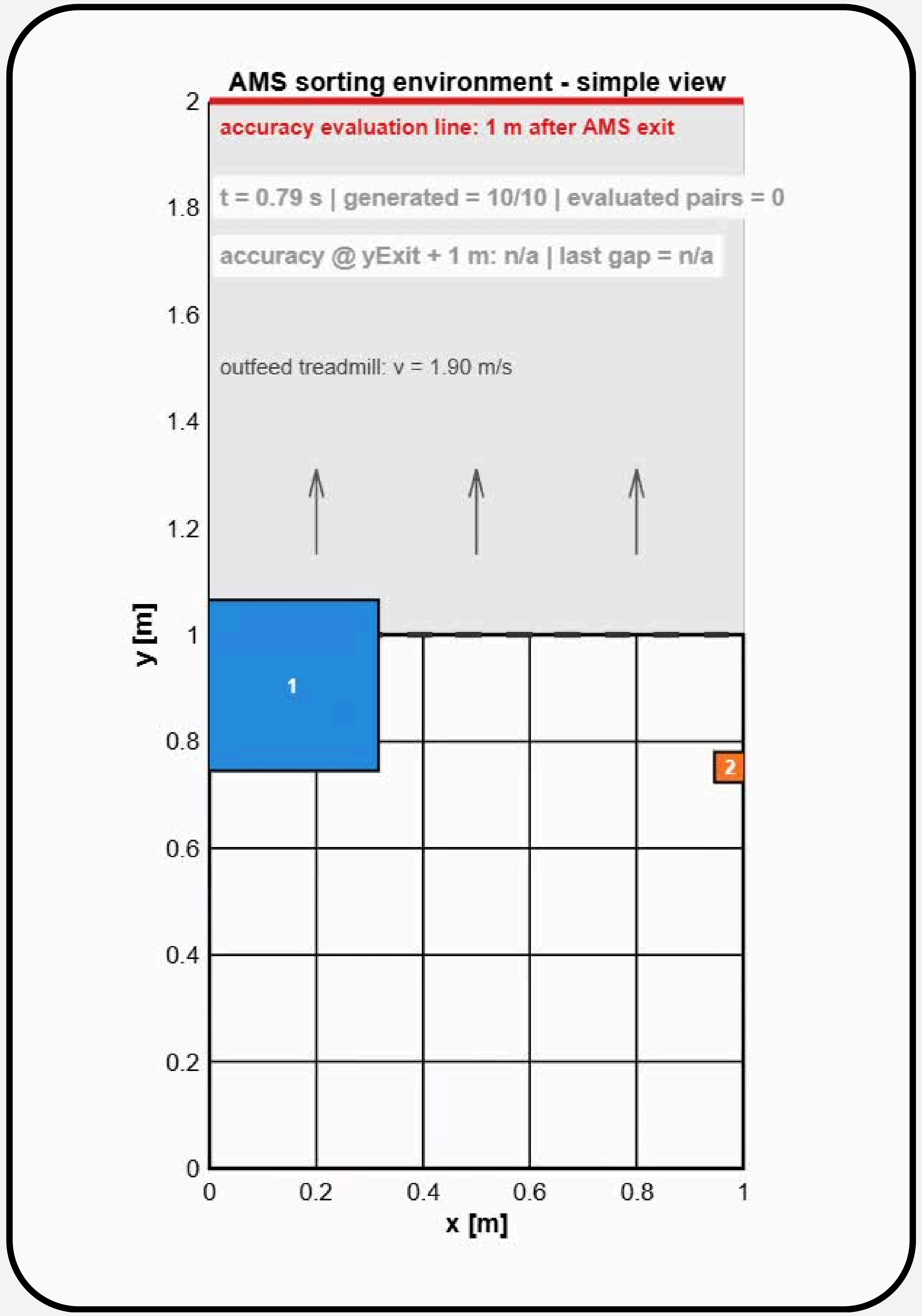
Sparse end-of-episode bonus when all 10 boxes cross the evaluation line with gap > 0. Provides long-horizon incentive. Prevents policies that work early but degrade near exit.

Pseudo-Collisions

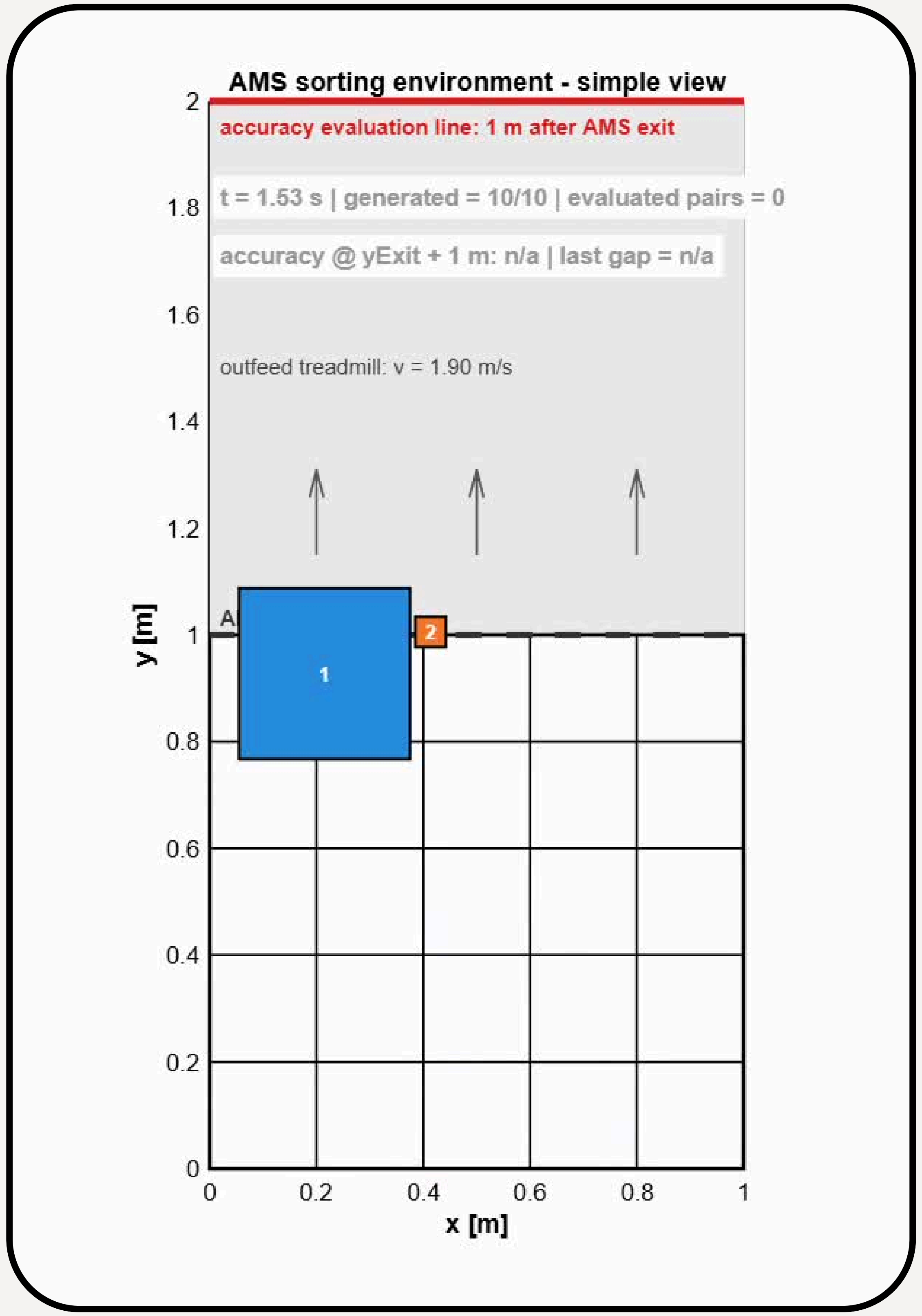
DIFF_LANE LOGIC

Problem: 1D (Y-axis) gap checks penalized safe X-lane overtakes, causing "Wall-hugging" reward hacking.
Solution: X-Lane Separation (diff_lane logic). Unlocked 2D mobility and 100% accuracy.

ABLATION STUDY

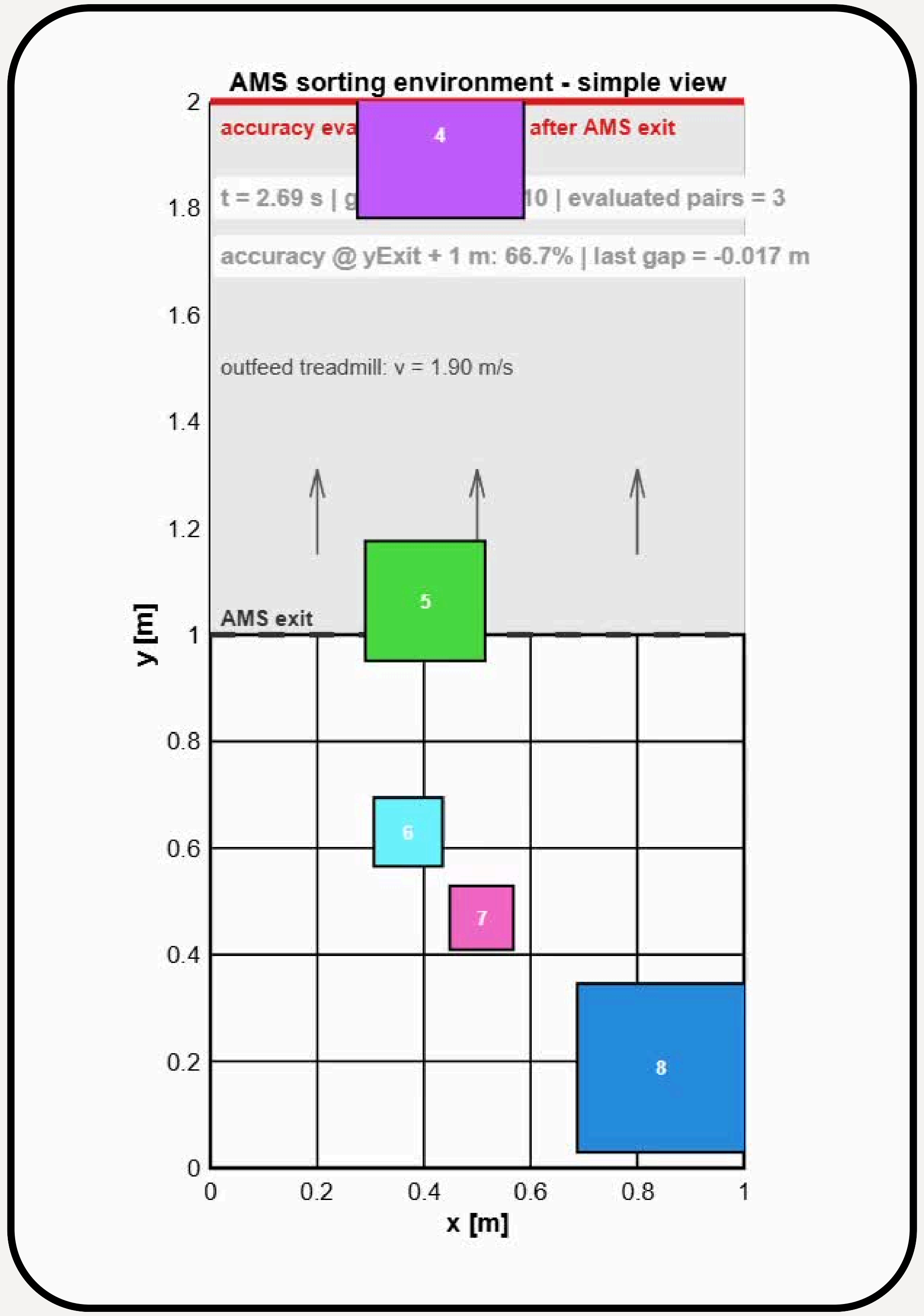


Training 1 (Wall-Hugging & Reward Hacking)

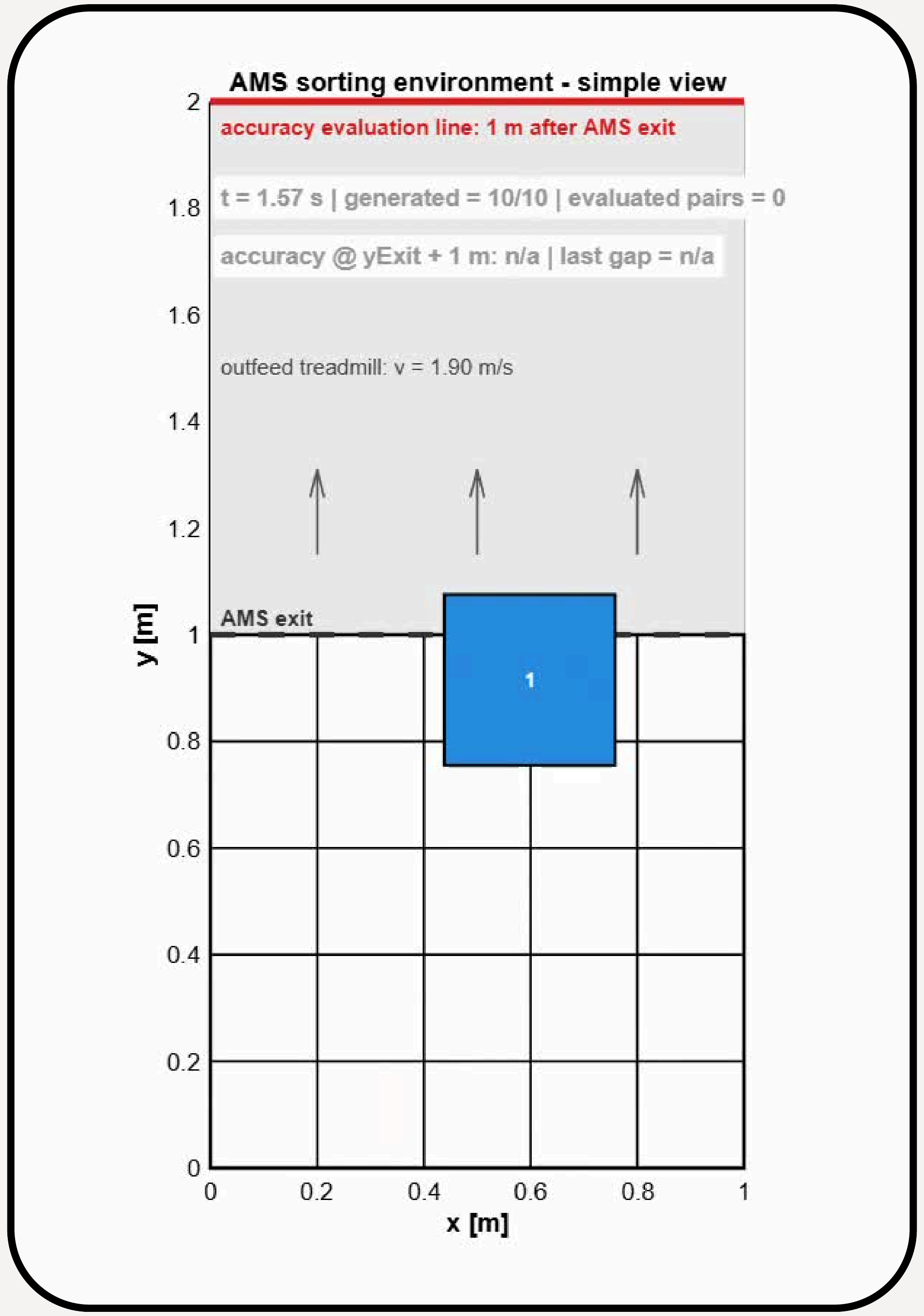


Training 5 (Over-penalization for centering & The Metric Illusion)

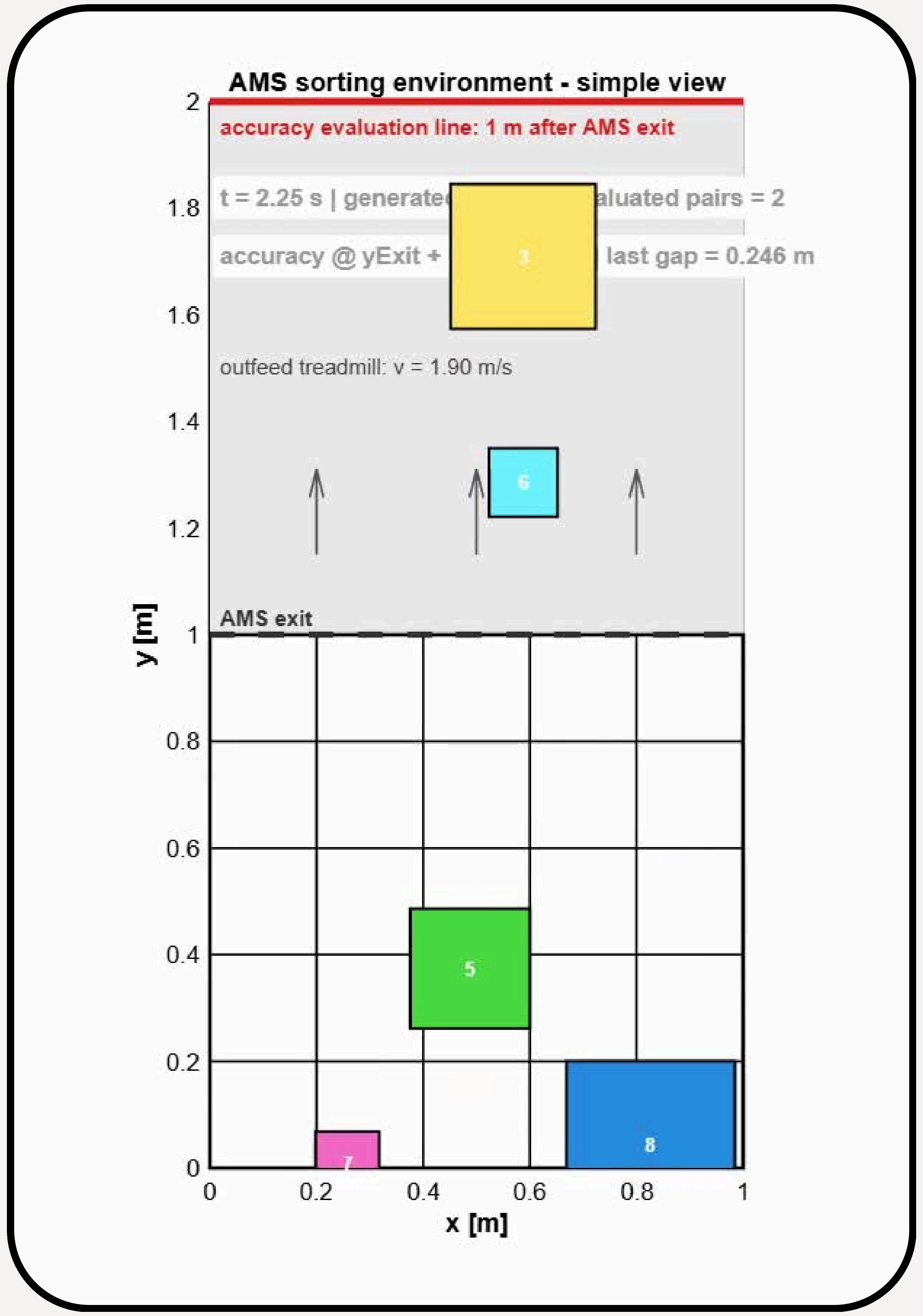
THE BASELINE: BEFORE VS. AFTER TRAINING



Untrained: Accuracy ~40% (Random), Avg. Gap: -0.128 m (Collision)



Trained (Run 11): Accuracy 100%, Avg Gap: 1.30 m (Safe Flow)



Trained (Run 11 with time based): Accuracy 100%, Avg Gap: 0.3 m (Safe Flow)

HYPERPARAMETER STRESS TEST

Set 1: Higher Entropy & Batch Size

agentOpts = rlPPOAgentOptions(...
ExperienceHorizon=1000, % Equalized to episode length.
ClipFactor=0.2, % PPO industry standard
EntropyLossWeight=0.02,...
MiniBatchSize=128, % RAM friendly learning packages
ActorOptimizerOptions=actorOpts,...
CriticOptimizerOptions=criticOpts,...
NumEpoch=3
AdvantageEstimateMethod="gae",...
GAEFactor=0.95,... & whether an action was “better”
SampleTime=0.01,...
DiscountFactor=0.99);

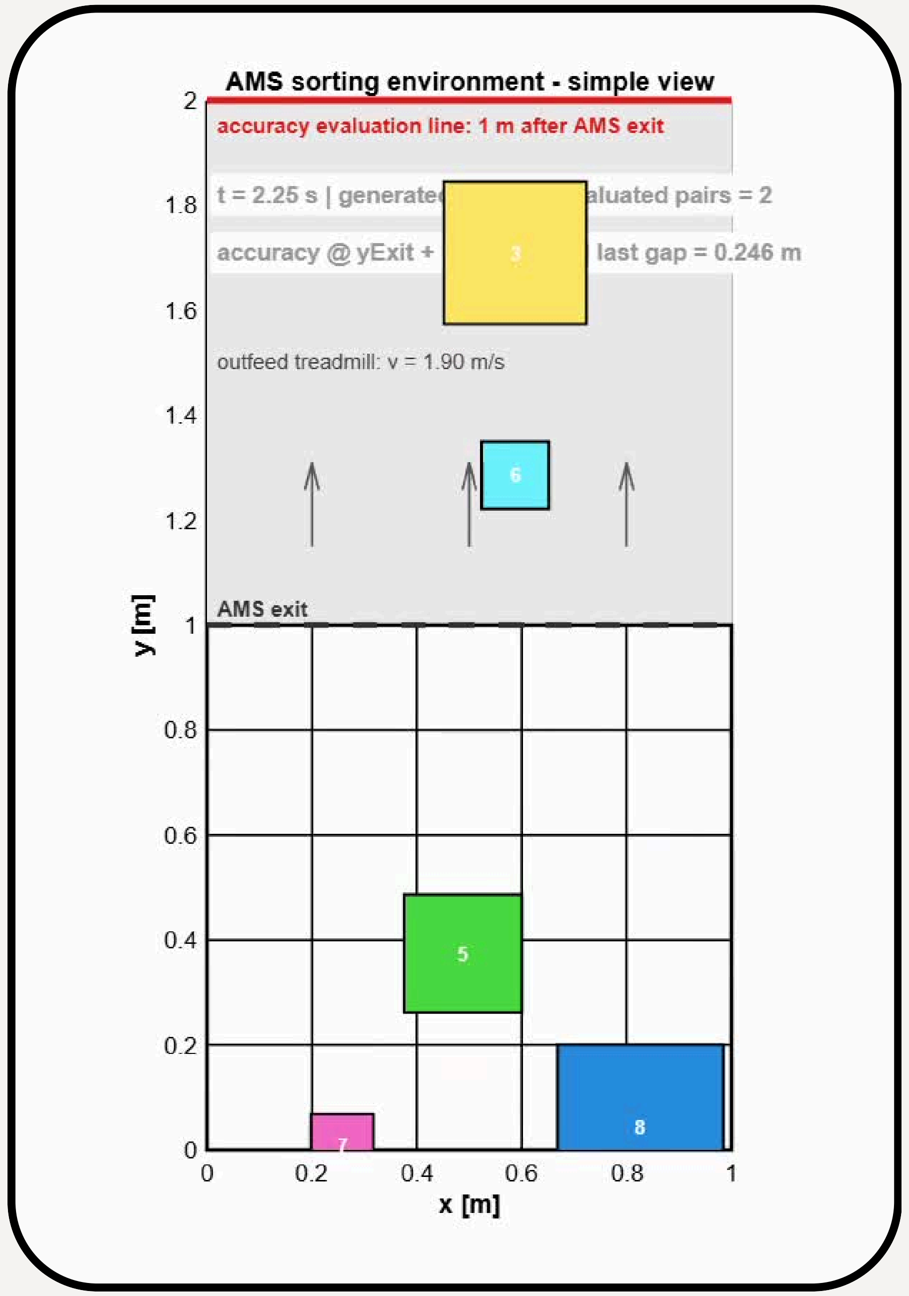
trainOpts = rlTrainingOptions(...
MaxEpisodes=25000,...
MaxStepsPerEpisode=1000,...
Plots="training-progress",...
StopTrainingCriteria="AverageReward",...
StopTrainingValue=40000,...
ScoreAveragingWindowLength=100,...
UseParallel=true);

Set 2: Lower Entropy & Batch Size

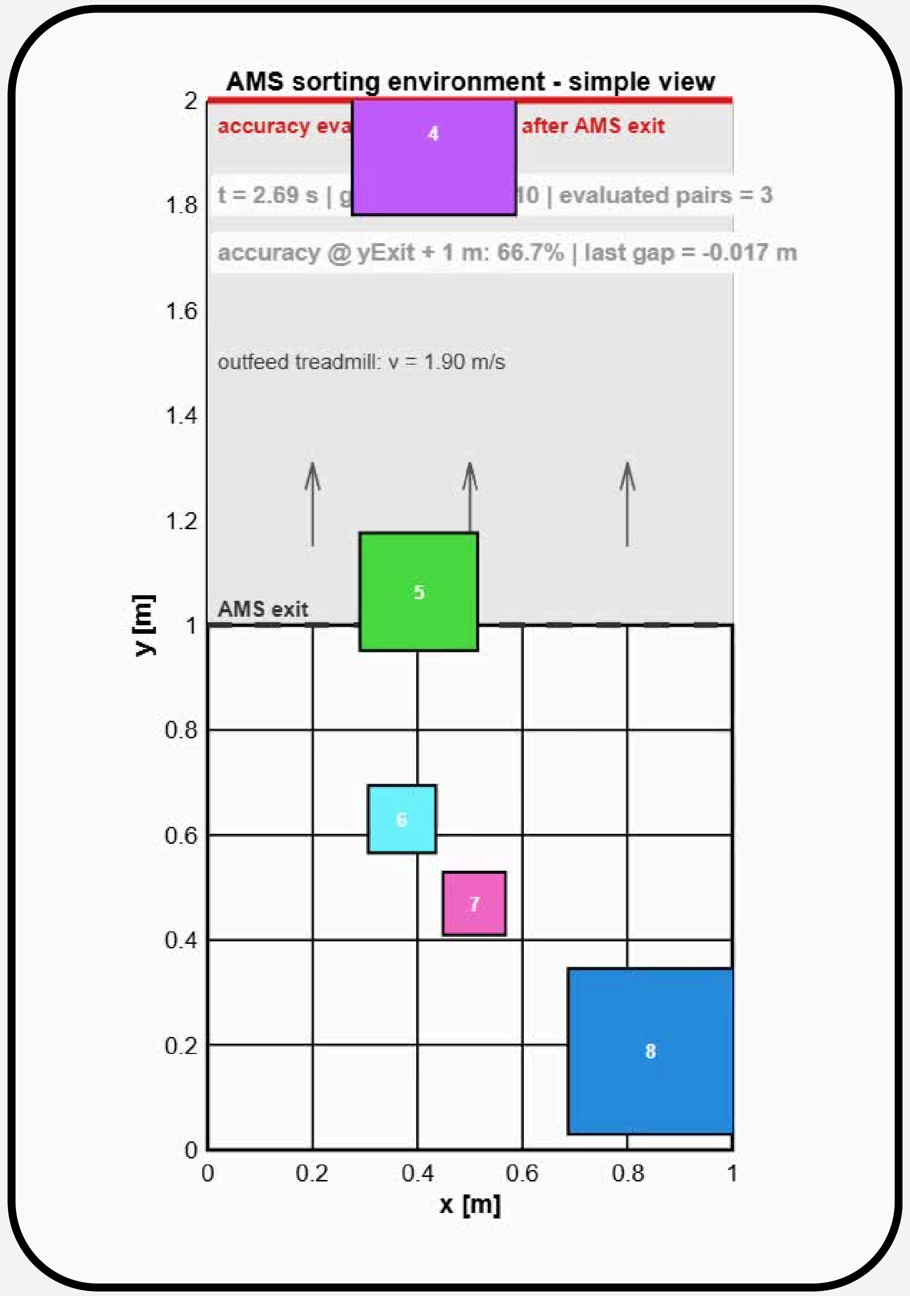
agentOpts = rlPPOAgentOptions(...
ExperienceHorizon=1000, % Equalized to episode length.
ClipFactor=0.2, % PPO industry standard
EntropyLossWeight=0.005,...
MiniBatchSize=64, % RAM friendly learning packages
ActorOptimizerOptions=actorOpts,...
CriticOptimizerOptions=criticOpts,...
NumEpoch=10
AdvantageEstimateMethod="gae",...
GAEFactor=0.95,...
SampleTime=0.01,...
DiscountFactor=0.99);

trainOpts = rlTrainingOptions(...
MaxEpisodes=25000,...
MaxStepsPerEpisode=1000,...
Plots="training-progress",...
StopTrainingCriteria="AverageReward",...
StopTrainingValue=40000,...
ScoreAveragingWindowLength=100,...
UseParallel=true);

HYPERPARAMETER STRESS TEST

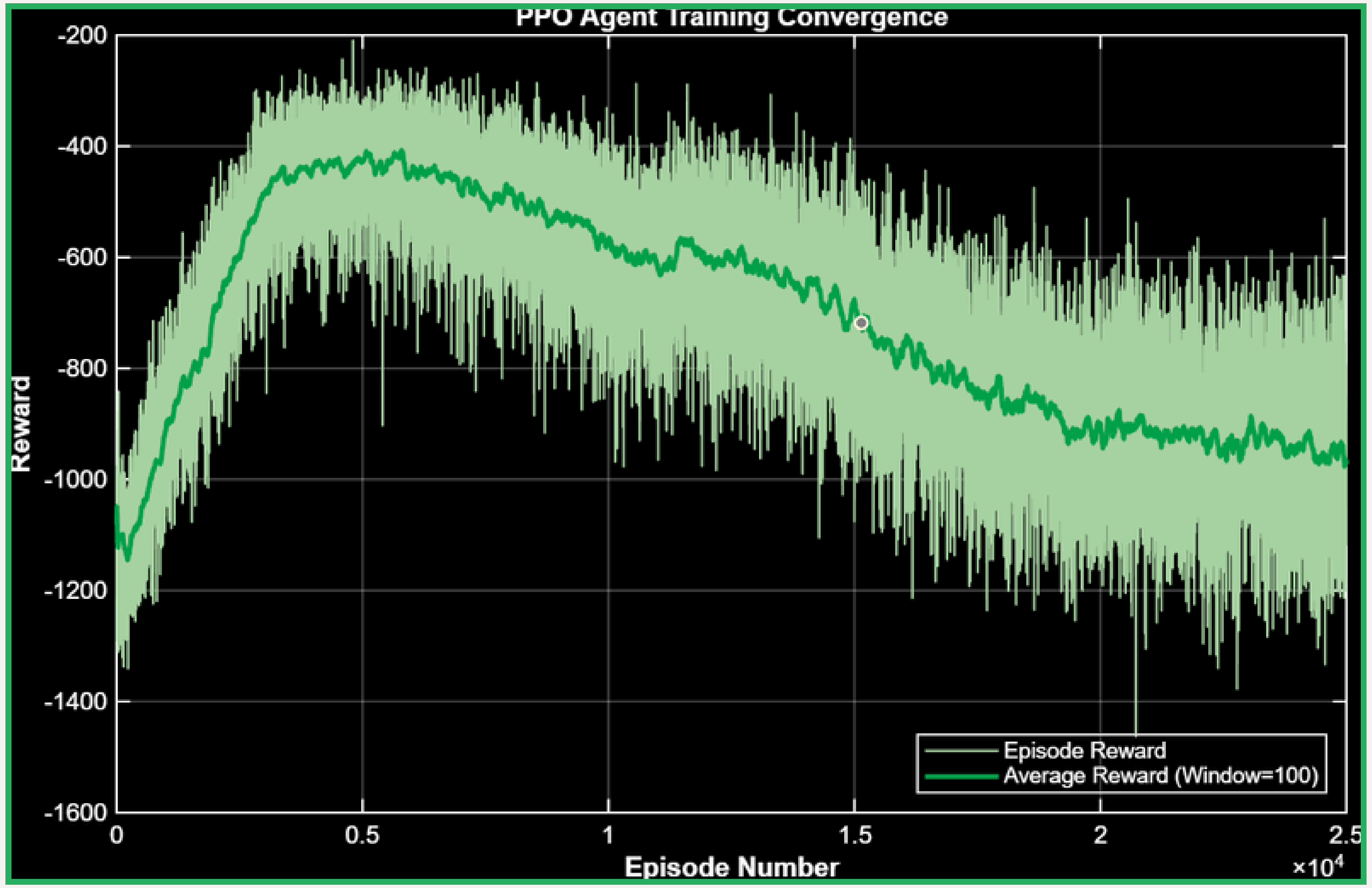


Baseline (Run 11 - Ideal):
Batch: 128 | Epoch: 3 | Entropy: 0.01
Result: Stable Convergence & 100% Accuracy.

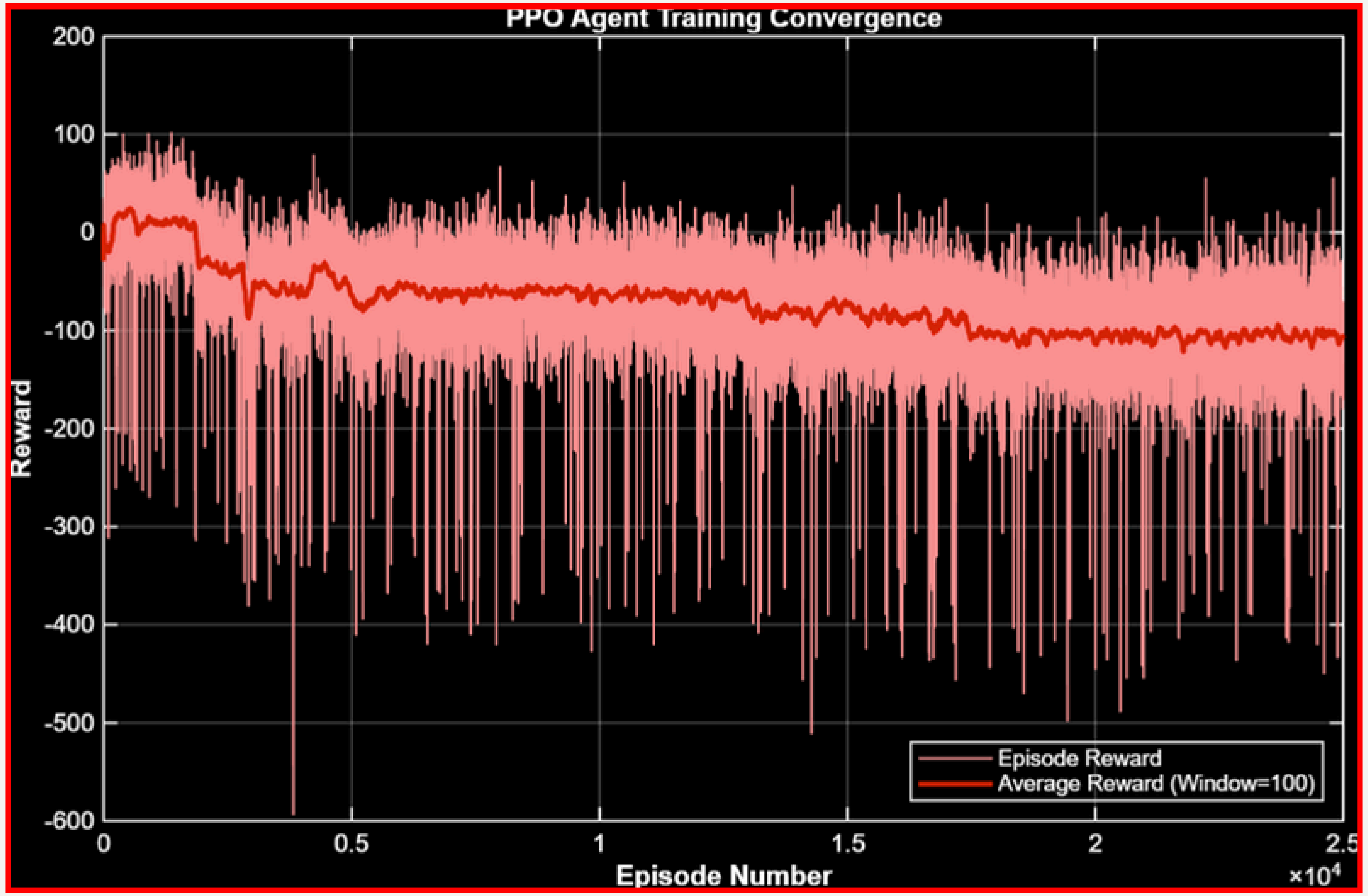


Stress Test (Failed):
Batch: 64 | Epoch: 10 | Entropy: 0.005
Result: Policy Collapse & Premature Convergence

HYPERPARAMETER STRESS TEST PLOTS



11th Training Reward Plot (Good)



11th Hyperparameter Training Reward Plot (Premature Convergence)

SUMMARY — RL PARCEL SINGULATION

Proof of Concept Success

The final RL policy (Run 11) successfully acts as a comprehensive 2D controller, achieving 100% accuracy and safe consolidation gaps (Y>1.3 m) in wave-based scenarios.

Throughput Bottleneck (Physical Limits)

Accuracy drops (to ~85%) in continuous 0.75 s stress tests are NOT algorithmic failures. They expose the strict hardware bottleneck of a 1.0 m^2 matrix and 1.9 m/s outfeed speed

Future Work (Sim-to-Real Gap)

The next engineering phase must bridge the "Sim-to-Real" gap by integrating actuator latency, sensor noise, and realistic friction dynamics into the training environment.

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THANK YOU